Infrastructure as Code (IaC) Specification

Declarative Provisioning for Guardian Streaming, Security & Persistence

The Guardian Layer DevOps Team

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1 Overview

This document outlines the Infrastructure as Code (IaC) design for the Guardian Layer using Terraform / OpenTofu. Declarative modules provision and manage the containerized message brokers, secrets managers, relational stores, and analytical observability engines underpinning the 3-layer urban monitoring platform.

2 Managed Infrastructure Modules

2.1 Streaming Broker Infrastructure

- **Apache Kafka Cluster (KRaft mode):** 3 broker nodes deployed across isolated network zones.
- **Pre-provisioned Topics:**
  - `telemetry-ingest` (alias `sensors.raw`): 12 partitions, replication factor 3.
  - `telemetry-validated` (alias `sensors.validated`): 12 partitions, replication factor 3.
  - `telemetry-dlq` (alias `sensors.dlq`): 6 partitions, 30-day retention for post-incident security forensics.
  - `devices.metadata`: 3 partitions with a compacted log cleanup policy.

2.2 Security Perimeter & Vault

- **HashiCorp Vault HA:** 3-node cluster utilizing Raft integrated storage with auto-unseal capabilities.
- **AppRole Automation:** Declarative provisioning of `guardian-app`, `twin-app`, `brain-app`, and `rotator` roles with scoped HCL policies.
- **Network Isolation:** Docker bridge networks isolating internal database listeners from public edge ports.

2.3 Persistence and Observability

- **PostgreSQL 16:** Managed relational instance for structured anomaly verdict tables, indexed temporal partitions, and connection pooling.
- **Elasticsearch 8.15:** 3-node cluster with Index Lifecycle Management (ILM) for 90-day forensic log archiving.

3 State Management & CI Deployment

```
terraform init
terraform plan -out=tfplan
terraform apply tfplan
```

State files reside in remote object storage with state locking enabled. Credentials and tokens are pulled dynamically from Vault, guaranteeing zero hard-coded secrets in repository sources.